

Optimal Growth Model.

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Kaldor Facts: ① ^{balanced growth} Real output growth = Constant. $\frac{Y_{t+1}}{Y_t} = \text{Const.}$

$\frac{I}{Y} = \text{Const.} \Leftrightarrow$ ② Stock of capital growth > Labor input growth. $\frac{K_{t+1}}{K_t} > \frac{L_{t+1}}{L_t}$

$\frac{rk}{Y} = \text{Const.} \Leftrightarrow$ ③ Real output growth = Stock of capital growth. $\frac{K_{t+1}}{K_t} = \frac{Y_{t+1}}{Y_t}$

④ Profit on capital = Constant. $r = \text{Const.}$

⑤ $\frac{\text{Profit}}{\text{Income}} \uparrow \cdot \frac{\text{Investment}}{\text{Output}} \uparrow$

⑥ output per capita growth varies across countries.

Ramsey - Cass - Koopmans Model.

Assumptions: Representative agent: $U'(\cdot) > 0$, $U''(\cdot) < 0$.

Production technology: $f'(\cdot) > 0$, $f''(\cdot) < 0$. Inada Condition

Social Planner: $V(k_0) = \max_{\{C_t, K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t U(C_t)$.

s.t. $C_t + K_{t+1} = f(K_t) + (1-\delta)K_t$

Bellman: $V(k) = \max_{k'} \{ U(f(k) + (1-\delta)k - k') + \beta V(k') \}$; $V'(\cdot) > 0$, $V''(\cdot) < 0$.

FOC: $\frac{\partial V}{\partial k'} = -U'(c) + \beta V'(k') = 0$. Suppose interior solution.

Envelope: $V'(k) = U'(c) [f'(k) + 1-\delta] \Rightarrow V'(k')$. Time invariant.

Euler: $U'(c) = \beta [f'(k') + 1-\delta] U'(c')$. Intertemporal substitution.

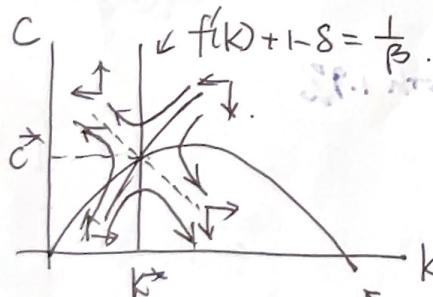
Transversality condition: $\lim_{t \rightarrow \infty} \beta^t U'(C_t) K_{t+1} = 0$. Complementary slackness condition.
 $\Leftrightarrow \lim_{t \rightarrow \infty} \beta^t U'(C_t) [f'(K_t) + 1-\delta] K_t = 0 \Rightarrow K_t K_{t+1} = \beta^t U'(C_t) K_{t+1} = 0$.

Steady state: $f'(k^*) = \frac{1}{\beta} + \delta - 1 = \rho + \delta$. $\rho := \frac{1}{\beta} - 1$

$\rho > 0$. $f'(k^*) > f'(k_{\text{golden}})$ in Solow-Swan Model.

Convergence: $k' = \alpha(k)$. 1st order expansion: $(k' - k^*) = \alpha'(k^*) (k - k^*)$.

Transition:



If $k_{t+1} < k^*$: $f'(k_{t+1}) > f'(k^*) \Rightarrow \frac{U'(C_t)}{U'(C_{t+1})} > 1 \Rightarrow C_{t+1} > C_t$.

If $C_{t+1} < f(K_t) - \delta K_t$: $k_{t+1} > k_t$.

Linearization around (k^*, c^*) : $(\Delta k_t, \Delta C_t) := (k_t - k^*, C_t - c^*)$

$\Delta C_{t+1} + \Delta k_{t+1} = \frac{1}{\beta} \Delta k_t$.

$\Delta C_{t+1} + f''(k^*) \Delta k_{t+1} \left[\frac{\beta U'(c^*)}{U''(c^*)} + \Delta C_{t+1} \right] = \Delta C_t$.

$c = f(k) - \delta k$.

Log-linearization $\begin{cases} f(x) - f(x^*) = f'(x^*) (x - x^*) \\ x - x^* = e^{\ln x} - e^{\ln x^*} = e^{\ln x^*} (\ln x - \ln x^*) \end{cases}$

RCL Model with Labor-Leisure Choice

Representative agent: $u(c, 1-h)$

$\nabla u > 0$ Jacobian
 $\nabla^2 u < 0$ Hessian

Production: $F(h, k)$

Bellman: $V(k) = \max_{\{h, k'\}} \{u(c, 1-h) + \beta V(k')\}$

s.t. $C = F(h, k) + (1-\delta)k - k'$

FOC: $\frac{\partial V}{\partial k'} = -u_1(c, 1-h) + \beta V'(k') = 0$ Intertemporal

$\frac{\partial V}{\partial h} = u_1(c, 1-h) F_h(h, k) - u_2(c, 1-h) = 0$ Intratemporal

Envelope: $V'(k) = u_1(c, 1-h) [F_k(h, k) + 1-\delta] \Rightarrow V'(k') = u_1(c', 1-h') [F_k(h', k') + 1-\delta]$

$\Rightarrow$ Euler: $u_1(c, 1-h) = \beta u_1(c', 1-h') [F_k(h', k') + 1-\delta]$

Intratemporal: $u_1(c, 1-h) F_h(h, k) = u_2(c, 1-h)$

Transversality: $\lim_{t \rightarrow \infty} \beta^t u_1(c_t, 1-h_t) k_{t+1} = 0$

Cobb-Douglas production: $F(h, k) = k^\theta h^{1-\theta}$: $F_k(h, k) = \frac{\theta Y}{k}$ $F_h(h, k) = \frac{(1-\theta)Y}{h} = (1-\theta) \left(\frac{k}{h}\right)^\theta$

$\Rightarrow$ Euler: $u_1(c, 1-h) = \beta u_1(c', 1-h') \left[\frac{\theta Y'}{k'} + 1-\delta\right]$

Intratemporal: $u_1(c, 1-h) \frac{(1-\theta)Y}{h} = u_2(c, 1-h)$

General utility: $u(c, 1-h) = \frac{[c^{1-\alpha} (1-h)^\alpha]^{1-\sigma} - 1}{1-\sigma}$

$u_1 = (1-\alpha) \frac{[c^{1-\alpha} (1-h)^\alpha]^{-\sigma}}{c} = \frac{(1-\alpha) Y^{1-\sigma}}{c}$
 $u_2 = \alpha \frac{[c^{1-\alpha} (1-h)^\alpha]^{-\sigma}}{1-h} = \frac{\alpha Y^{1-\sigma}}{1-h}$
 $IES := \frac{-d \ln(\frac{c_{t+1}}{c_t})}{d \ln R_{t+1}} = \frac{-d \ln(\frac{c_{t+1}}{c_t})}{d \ln(\frac{u'(c_{t+1})}{u'(c_t)})}$
 $= \frac{-d \ln c_t}{d \ln u'(c_t)} = \frac{-u'(c)}{c u''(c)} = \frac{1}{RRA}$

IES: only affects the path, but not steady state.

$\sigma \rightarrow 1$: Simplified utility $u(c, 1-h) = \ln c + \frac{\alpha}{1-\alpha} \ln(1-h)$: $u_1 = \frac{1}{c}$ $u_2 = \frac{\alpha}{1-\alpha} \frac{1}{1-h}$

$\Rightarrow$ Euler: $\frac{1}{c} = \beta \frac{1}{c'} \left[\frac{\theta Y'}{k'} + 1-\delta\right]$

Intratemporal: $\frac{1}{c} \frac{(1-\theta)Y}{h} = \frac{\alpha}{1-\alpha} \frac{1}{1-h} =: A \frac{1}{1-h}$

Steady state: $\frac{\theta Y^*}{k^*} = \frac{1}{\beta} - 1 + \delta = \rho + \delta \Rightarrow \frac{k^*}{Y^*} = \frac{\theta}{\rho + \delta}$, $\frac{\dot{Y}^*}{Y^*} = -\frac{\theta \delta}{\rho + \delta}$

$\left\{ \begin{array}{l} \frac{(1-\theta)Y^*}{h^*} = \frac{A c^*}{1-h^*} \\ c^* = Y^* - \delta k^* \end{array} \right\} \Rightarrow \frac{h^*}{1-h^*} = \frac{(1-\theta)Y^*}{A(Y^* - \delta k^*)} \Rightarrow h^* = \frac{1-\theta}{1-\theta + A(1-\delta \frac{k^*}{Y^*})}$

Calibration: Population growth 1.5% exogenous technological growth 1.9%

① labor income share = 0.7

$\Rightarrow \theta = 0.3$

$\Rightarrow \delta: \frac{k}{Y} = \frac{\theta}{\rho + \delta}$ $\delta = 0.1$

② Employment share of discretionary time = $\frac{1}{3}$

$\Rightarrow h = \frac{1}{3}$

③ Annualized interest rate = 5%

$\Rightarrow \boxed{\rho = 0.05}$

④ Annual capital output ratio = 2

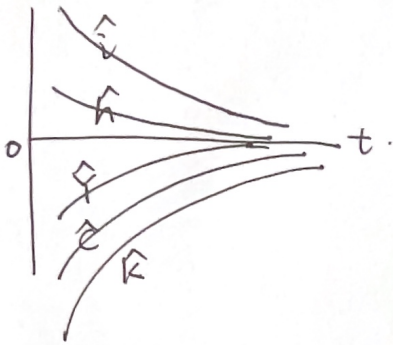
$\Rightarrow \frac{k}{Y} = 2$

$A: h = \frac{1-\theta}{1-\theta + A(1-\delta \frac{k}{Y})}$

$A = 1.75$

(Durable) stock $\uparrow$ flow

Transition. $\hat{x} := \ln \frac{x_t}{x^*}$



$$\frac{(1-\theta)Y_t}{C_t} = \frac{A h_t}{1-h_t} \Rightarrow \hat{Y}_t = \hat{C}_t + \hat{h}_t - \hat{h}_t > \hat{C}_t$$

$$C_{t+1} = \beta C_t \left[\frac{\theta Y_{t+1}}{K_{t+1}} + 1 - \delta \right] \Rightarrow \hat{C}_t = \hat{C}_{t+1} - \beta(1+\delta)(\hat{Y}_{t+1} - \hat{K}_{t+1})$$

$$C_t + K_{t+1} = Y_t + (1-\delta)K_t \Rightarrow \left(\frac{1+\delta}{\theta} - \delta\right) \hat{C}_t + \hat{K}_{t+1} = \frac{1+\delta}{\theta} \hat{Y}_t + (1-\delta) \hat{K}_t$$

$$Y_t = K_t^\theta h_t^{1-\theta} \Rightarrow \hat{Y}_t = \theta \hat{K}_t + (1-\theta) \hat{h}_t$$

$$\dot{K}_t = K_{t+1} - (1-\delta)K_t \Rightarrow \hat{K}_t = \delta \hat{K}_{t+1} - \delta(1-\delta) \hat{K}_t$$

With endogenous labor, capital stock converges faster to SS.

Transition speed $\uparrow$: Capital share $\downarrow$.

IES. $\sigma \uparrow$.

Capital adjustment cost $\downarrow$.

Recursive Competitive Equilibrium (Prescott & Mehra 1980).

A RCE is a list of functions of $x = k$, denoted by $\{K'(x), H(x), w(x), r(x), k(k, x), h(k, x), v(k, x)\}$ s.t.

1. Given $\{w(k), r(k)\}, \{H(k), K\}$ solves firm's profit maximization problem:

$$\max_{\{H, K\}} F(H, K) - wH - rK \Rightarrow \text{FOC} \begin{cases} F_H(H, K) = w(k) \\ F_K(H, K) = r(k) \end{cases}$$

$$K'(k)$$
2. Given $\{w(k), r(k)\}, \{K'(k, k), h(k, k)\}$ solves consumer's utility maximization problem, generating the value function $v(k, k)$.

$$v(k, k) = \max_{\{h, k'\}} \{u(c) + \beta v(k', k')\} \Rightarrow \text{Euler: } u'(c) = \beta u'(c') [r(k') + 1 - \delta]$$

$$\text{s.t. } c + k' = w(k)h + r(k)k + (1-\delta)k$$

$$K' = K'(k)$$

$$\text{TVC: } \lim_{t \rightarrow \infty} \beta^t u'(c_t) K_{t+1} = 0$$

$$\text{LOM: } k' = w(k)h + r(k)k + (1-\delta)k - c$$

$$k' = K'(k)$$
3. Market clearing (Consistency): Labor: $h(k, k) = H(k)$.
 "Rational belief" Capital: $K'(k, k) = K'(k)$.

Without distortion, RCE $\Leftrightarrow$ Social planner's allocation.

With distortions: ① taxes, inflation, tariffs.

② externalities: pollution, congestion, network effects, knowledge spillovers

With taxes: τ_h, τ_k, T .

Social planner: $\max_{\{C_t, H_t, k_{t+1}, G_t\}_{t=0}^{+\infty}} \sum_{t=0}^{+\infty} \beta^t u(C_t, 1-H_t)$
 s.t. $C_t + I_t + G_t = F(H_t, k_t)$
 $k_{t+1} = (1-\delta) k_t + I_t$

Bellman: $V(k) = \max_{\{H, G, k'\}} \{u(C, 1-H) + \beta V(k')\}$
 s.t. $C = F(H, k) + (1-\delta)k - G - k'$

Optimal policy function $H(k), G(k), k'(k)$ & value function $V(k)$ satisfy:

$$\begin{cases} \text{FOC: } u_1(C(k), 1-H(k)) = \beta V'(k') & \text{on } k' \\ u_2(C(k), 1-H(k)) = u_1(C(k), 1-H(k)) \cdot F_H(H(k), k) & \text{on } H \\ G(k) = 0 & \text{on } G \\ \text{Envelope: } V'(k) = u_1(C(k), 1-H(k)) [F_k(H(k), k) + 1-\delta] \end{cases}$$

RCE: A list of functions

1. Given $\{w(k), r(k)\}$, $\{H(k), k\}$ solves firm's profit maximization problem.

$$\max_{H, k} F(H, k) - wH - rk \quad \text{FOC: } \begin{cases} w(k) = F_H(H(k), k) \\ r(k) = F_k(H(k), k) \end{cases}$$

2. Given $\{w(k), r(k), \tau_h, \tau_k, T(k)\}$, $\{h(k, k), k(k, k)\}$ solves consumer's uwp, generating value function $v(k, k)$.

$$v(k, k) = \max_{\{h, k'\}} \{u(C, 1-h) + \beta v(k', k')\}$$

$$\text{s.t. } C = w(k)h(1-\tau_h) + r(k)k(1-\tau_k) + (1-\delta)k - k' - T(k)$$

$$k' = k(k)$$

$$\Rightarrow \begin{aligned} u_1(C(k, k), 1-h(k, k)) &= \beta \frac{\partial v}{\partial k'}(k', k') \\ u_2(C(k, k), 1-h(k, k)) &= u_1(C(k, k), 1-h(k, k)) w(k) (1-\tau_h) \\ \frac{\partial v}{\partial k}(k, k) &= u_1(C(k, k), 1-h(k, k)) [r(k)(1-\tau_k) + 1-\delta] \end{aligned}$$

3. Given $\{w(k), r(k), H(k), k\}$, $\{\tau_h, \tau_k, T(k)\}$ satisfies government's tight budget.
 $G(k) = w(k)H(k)\tau_h + r(k)k\tau_k + T(k)$

4. Markets clear (Consistency): Labor: $h(k, k) = H(k)$
 Capital: $k(k, k) = k(k)$

$$\Rightarrow \begin{cases} u_1(C(k), 1-H(k)) = \beta V'(k') \\ u_2(C(k), 1-H(k)) = u_1(C(k), 1-H(k)) F_H(H(k), k) (1-\tau_h) \\ V'(k) = u_1(C(k), 1-H(k)) [F_k(H(k), k) (1-\tau_k) + 1-\delta] \\ C(k) = F(H(k), k) + (1-\delta)k - k(k) - G(k) \end{cases}$$

RCE $\Leftrightarrow$ SPA.

iff. $\tau_h = \tau_k = 0$

Assumption: $u(c, 1-h) = \ln c + A \ln(1-h)$.

$$F(H, K) = K^\theta H^{1-\theta} =: Y$$

$$Y^* = \left(\frac{K^*}{Y^*}\right)^{\frac{\theta}{1-\theta}} H^*, \quad C^* = (1-\delta - \theta \frac{K^*}{Y^*}) Y^*.$$

Steady state: $\frac{\partial Y^*}{K^*} = \frac{\rho + \delta}{1-\tau_k} \Rightarrow \frac{K^*}{Y^*} = \frac{\theta(1-\tau_k)}{\rho + \delta}$

$$\tau_k \uparrow \cdot \frac{K^*}{Y^*} \downarrow$$

$$\frac{A}{1-H^*} = \frac{1}{C^*} \frac{(1-\theta)Y^*}{H^*} (1-\tau_h) \Rightarrow H^* = \frac{1-\theta}{1-\theta + \frac{A}{1-\tau_h} (1-\delta - \theta \frac{K^*}{Y^*})} \quad \text{If } G = \delta Y.$$

$$C^* = Y^* - \delta K^* - G^*.$$

$$\begin{aligned} \tau_k \uparrow &\cdot H^* \downarrow \cdot \frac{K^*}{Y^*} \downarrow \cdot Y^* \downarrow \cdot K^* \downarrow \cdot C^* \downarrow \cdot r^* \uparrow \cdot w^* \downarrow \\ \tau_h \uparrow &\cdot H^* \downarrow \cdot \frac{K^*}{Y^*} \downarrow \cdot Y^* \downarrow \cdot K^* \downarrow \cdot C^* \downarrow \cdot r^* \downarrow \cdot w^* \downarrow \\ \delta \uparrow &\cdot H^* \uparrow \cdot \frac{K^*}{Y^*} \downarrow \cdot Y^* \uparrow \cdot K^* \uparrow \cdot C^* \downarrow \cdot r^* \downarrow \cdot w^* \downarrow \end{aligned}$$

If $G = \bar{G}$ fixed: $H^* = \frac{1-\theta}{1-\theta + \frac{A}{1-\tau_h} (1-\bar{G} - \delta \frac{K^*}{Y^*})}$

$$Y^* = \left(\frac{K^*}{Y^*}\right)^{\frac{\theta}{1-\theta}} H^*.$$

$$\Rightarrow H^* = \frac{1-\theta + \frac{A}{1-\tau_h} \bar{G} \left(\frac{K^*}{Y^*}\right)^{\frac{\theta}{1-\theta}}}{1-\theta + \frac{A}{1-\tau_h} (1-\delta \frac{K^*}{Y^*})} = 1 - \frac{-\bar{G} \left(\frac{K^*}{Y^*}\right)^{\frac{\theta}{1-\theta}} + 1 - \delta \frac{K^*}{Y^*}}{\frac{1}{A} (1-\theta) (1-\tau_h) + 1 - \delta \frac{K^*}{Y^*}}.$$

$$\tau_k \uparrow \quad H^* \downarrow$$

$$\tau_h \uparrow \quad H^* \downarrow$$

$$\bar{G} \uparrow \quad H^* \uparrow$$

In the long run: $\tau_k = 0$. $\tau_h > 0$ is optimal since K is more elastic.
In dynamic prog.: $\tau_k > 0$. Since at each t , K_t is inelastically given.

Taxing in labor is better than capital: ① capital income < labor income at τ . ② Capital is more elastic to tax increase.

Dependence in taxes: $\tau_h(\tau_k) = \frac{G - rK\tau_k}{WH} = \frac{\delta - \theta\tau_k}{1-\theta}$

$$H^* = \frac{1}{1 + A \frac{1-\delta - \frac{\delta\theta}{\rho+\delta} (1-\tau_k)}{1-\delta - \theta(1-\tau_k)}}$$

$$\tau_k \uparrow \cdot \underline{H^* \uparrow} \cdot \frac{K^*}{Y^*} \downarrow \cdot Y^* \downarrow \cdot K^* \downarrow \cdot C^* \downarrow \cdot r^* \uparrow \cdot w^* \downarrow.$$

Effect of τ_k on H^* or K^* depends on ① preference (elasticity).

② government spending. $G = \delta Y$ or $G = \bar{G}$

③ agents' expectation on taxes.

Effect of τ_k on Y^* : $\frac{Y(\tau_k)}{Y(0)} = \left(\frac{1-\tau_k}{1-\tau_k^0}\right)^{\frac{\theta}{1-\theta}}$

Labor supply: $u(c, 1-h) = \ln c + \frac{A}{1-\eta} [(1-h)^{1-\eta} - 1] \Rightarrow \text{FOC: } u_c w = A(1-h)^{-\eta}.$

"Intensive Margin"

$$\Rightarrow \text{log-linearization: } \hat{u}_c + \hat{w} = -\eta \cdot \hat{1-h}$$

Budget: $h + (1-h) = 1 \Rightarrow \text{log-linearization: } h \hat{h} + (1-h) \hat{1-h} = 0.$

$$\Rightarrow \hat{h} = \frac{1-h}{h} \frac{1}{\eta} (\hat{u}_c + \hat{w}). \quad \text{Frisch elasticity of labor supply: } \epsilon := \frac{\hat{h}}{\hat{w}} = \frac{1-h}{h} \frac{1}{\eta}$$

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Cobb-Douglas: $U(c, h) = \frac{c^{1-\sigma} (1 - \alpha(hc)^b)^{1-\sigma} - 1}{1-\sigma}$

CRRA utility (RCK Model)

Representative agent: $u(c) = \frac{c^{1-\varepsilon} - 1}{1-\varepsilon}$ $\varepsilon > 0$

Production: $F(k) = Ak$

Bellman: $V(k) = \max_{k'} \{ u(c) + \beta V(k') \}$

Define $B := A + 1 - \delta$

s.t. $c = Ak + (1-\delta)k - k' \Rightarrow c = Bk - k'$ Budget.

FOC: $\frac{\partial V}{\partial k'} = -u'(c) + \beta V'(k') = 0$ $u'(c) = c^{-\varepsilon}$

Envelope: $V'(k) = \beta u'(c)$ $\Rightarrow V'(k') = \beta u'(c')$

Euler: $u'(c) = \beta \beta u'(c')$ $\Rightarrow c^{-\varepsilon} = \beta \beta c'^{-\varepsilon} \Rightarrow c_t^{\varepsilon} = (\beta \beta)^t c_0^{\varepsilon}$ ①

Transversality: $\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0$ $\lim_{t \rightarrow \infty} \beta^t c_t^{-\varepsilon} k_{t+1} = 0$ ②

Method 1: Solve for 2nd differential equation

$$\textcircled{1} \Leftrightarrow \frac{B k_{t+1} - k_{t+2}}{B k_t - k_{t+1}} = (\beta \beta)^{\frac{1}{\varepsilon}} \Leftrightarrow \frac{B \frac{k_{t+1}}{k_{t+2}} - 1}{B \frac{k_t}{k_{t+1}} - 1} = (\beta \beta)^{\frac{1}{\varepsilon}} \frac{k_{t+1}}{k_{t+2}}$$

Define $X_t := \frac{k_t}{k_{t+1}} \in [\frac{1}{B}, +\infty)$ $\textcircled{1} \Leftrightarrow \frac{B X_{t+1} - 1}{B X_t - 1} = (\beta \beta)^{\frac{1}{\varepsilon}} X_{t+1}$

$$\Leftrightarrow \frac{1}{B X_t - 1} = \frac{(\beta \beta)^{\frac{1}{\varepsilon}}}{B} + \frac{(\beta \beta)^{\frac{1}{\varepsilon}}}{B} \frac{1}{B X_{t+1} - 1}$$

$$\Leftrightarrow \frac{1}{B X_{t+1} - 1} - \frac{1}{\frac{B}{(\beta \beta)^{\frac{1}{\varepsilon}}} - 1} = \frac{B}{(\beta \beta)^{\frac{1}{\varepsilon}}} \left[\frac{1}{B X_t - 1} - \frac{1}{\frac{B}{(\beta \beta)^{\frac{1}{\varepsilon}}} - 1} \right]$$

$$\Leftrightarrow \frac{1}{B X_t - 1} = \frac{1}{\frac{B}{(\beta \beta)^{\frac{1}{\varepsilon}}} - 1} + \left[\frac{B}{(\beta \beta)^{\frac{1}{\varepsilon}}} \right]^t \left[\frac{1}{B X_0 - 1} - \frac{1}{\frac{B}{(\beta \beta)^{\frac{1}{\varepsilon}}} - 1} \right] \text{ Unstable Fixed point.}$$

$$\textcircled{2} \lim_{t \rightarrow \infty} \beta^t \frac{k_{t+1}}{(B k_t - k_{t+1})^{\varepsilon}} = \lim_{t \rightarrow \infty} \beta^t \frac{k_{t+1}^{1-\varepsilon}}{(B X_t - 1)^{\varepsilon}} < +\infty$$

$$\textcircled{2.1} \Rightarrow \frac{1}{B X_t - 1} = \frac{1}{\frac{B}{(\beta \beta)^{\frac{1}{\varepsilon}}} - 1} \Rightarrow X_t = (\beta \beta)^{\frac{1}{\varepsilon}} \Rightarrow k_{t+1} = (\beta \beta)^{\frac{1}{\varepsilon}} k_t \Rightarrow k_t = (\beta \beta)^{\frac{t}{\varepsilon}} k_0$$

$$\textcircled{2.2} \Rightarrow \lim_{t \rightarrow \infty} \beta^t k_{t+1}^{1-\varepsilon} = 0 : \lim_{t \rightarrow \infty} \beta^t (\beta \beta)^{\frac{t(1-\varepsilon)}{\varepsilon}} k_0^{1-\varepsilon} = \lim_{t \rightarrow \infty} \left(\beta^{\frac{1}{\varepsilon}} B^{\frac{1-\varepsilon}{\varepsilon}} \right)^t k_0^{1-\varepsilon} = 0$$

$$\Leftrightarrow \beta^{\frac{1}{\varepsilon}} B^{\frac{1-\varepsilon}{\varepsilon}} < 1 \Leftrightarrow (\beta \beta)^{\frac{1}{\varepsilon}} < B \text{ Interior Solution}$$

Method 2: Guess & verify (Guess from Euler & TVC)

$$\text{Guess } k_{t+2} = \gamma k_{t+1} : \frac{(B - \gamma) k_{t+1}}{B k_t - k_{t+1}} = (\beta \beta)^{\frac{1}{\varepsilon}} \Leftrightarrow k_{t+1} = \frac{B}{(B - \gamma)(\beta \beta)^{\frac{1}{\varepsilon}} + 1} k_t$$

$$\text{Same } \gamma : \gamma = \frac{B}{(B - \gamma)(\beta \beta)^{\frac{1}{\varepsilon}} + 1} \Leftrightarrow (\gamma - B)(\gamma - (\beta \beta)^{\frac{1}{\varepsilon}}) = 0 \Leftrightarrow \gamma_1 = B, \gamma_2 = (\beta \beta)^{\frac{1}{\varepsilon}} \xleftarrow{\text{TVC}}$$

Finite Horizon Dynamic / Sequential Problem.

$$\max_{\{G_t, K_{t+1}\}_{t=0}^T} \sum_{t=0}^T \beta^t u(G_t) \quad \text{s.t.} \quad G_t + K_{t+1} = f(K_t) + (1-\delta)K_t.$$

$$L = \sum_{t=0}^{T-1} \beta^t \left\{ u(G_t) + \pi_t [f(K_t) + (1-\delta)K_t - G_t - K_{t+1}] \right\}.$$

$$\frac{\partial L}{\partial G_t} = \beta^t (u'(G_t) - \pi_t) = 0.$$

$$\frac{\partial L}{\partial K_{t+1}} = \beta^t (-\pi_t + \pi_{t+1} (f'(K_{t+1}) + 1 - \delta)) = 0.$$

Complementary slackness condition: $\pi_t [f(K_t) + (1-\delta)K_t - G_t - K_{t+1}] = 0.$

$$L = \sum_{t=0}^T \beta^t \left\{ u(f(K_t) + (1-\delta)K_t - K_{t+1}) \right\} + \pi_T K_{T+1}$$

$$\frac{\partial L}{\partial K_{t+1}} = -\beta^t u'(G_t) + \beta^t \pi_t + \beta^{t+1} u'(G_{t+1}) [f'(K_{t+1}) + 1 - \delta] = 0$$

$$\frac{\partial L}{\partial K_{T+1}} = -\beta^T u'(G_T) + \pi_T = 0 \Rightarrow \pi_T = \beta^T u'(G_T)$$

Complementary slackness condition: $\pi_T K_{T+1} = 0$
 $\Leftrightarrow \beta^T u'(G_T) K_{T+1} = 0$

Lagrangian & Bellman.

(HJB) Discrete time

$$V(x) = \max_{c, x'} \{ u(c, x) + \beta V(x') \}$$

$$\text{s.t. } x' = f(c, x)$$

$$L(x) = u(c, x) + \beta V(x') + \pi [f(c, x) - x']$$

$$\text{FOC: } \frac{\partial L}{\partial c} = u_1(c, x) + \pi f_{11}(c, x) = 0 \Rightarrow \pi = - \frac{u_1(c, x)}{f_{11}(c, x)}$$

$$\frac{\partial L}{\partial x'} = \beta V'(x') - \pi = 0$$

$$\Rightarrow \beta u_2(c', x') + [\beta f_{21}(c', x') - 1] \pi = 0$$

$$\text{Envelope: } V'(x) = u_2(c, x) + \pi f_{21}(c, x)$$

$$\Rightarrow \beta u_2(c', x') = \frac{u_1(c, x)}{f_{11}(c, x)} [\beta f_{21}(c', x') - 1]$$

$$\Rightarrow V'(x) = u_2(c', x') + \pi f_{21}(c', x')$$

Envelope condition

Transition: ① set up Euler + Budget + TVC.

② Intuition: Convergence to SS.

③ Intuition: Convergence to SS.

Brock & Mirman 1972. (RCK Model)

Representative agent: $u(C) = \ln C$

Production: $F(k) = k^\theta$

Depreciation: $\delta = 1$

Bellman: $V(k) = \max_{k'} \{ \ln(k^\theta - k') + \beta V(k') \} \quad k' \in [0, k^\theta]$

FOC: $\frac{\partial V}{\partial k'} = \frac{-1}{k^\theta - k'} + \beta V'(k') = 0$

Envelope: $V'(k) = \frac{\theta k^{\theta-1}}{c} \Rightarrow V'(k') = \frac{\theta k'^{\theta-1}}{c'}$

Euler: $\frac{1}{c} = \beta \frac{1}{c'} \theta k'^{\theta-1} \Rightarrow \frac{k_{t+1}^\theta - k_{t+2}^\theta}{k_t^\theta - k_{t+1}^\theta} = \beta \theta k_{t+1}^{\theta-1} \quad (1)$

Transversality: $\lim_{t \rightarrow \infty} \beta^t u'(C_t) k_{t+1} = 0$. (2) Transversality $\Leftrightarrow$ Interior solution condition

(1) $\Leftrightarrow \frac{\frac{k_{t+1}^\theta}{k_{t+2}^\theta} - 1}{\frac{k_t^\theta}{k_{t+1}^\theta} - 1} = \beta \theta \frac{k_{t+1}^\theta}{k_{t+2}^\theta}$ Method 1. Solve for 2nd differential equation not needed

Define $X_t := \frac{k_t^\theta}{k_{t+1}^\theta} \in [1, +\infty)$: (1) $\Leftrightarrow \frac{X_{t+1} - 1}{X_t - 1} = \beta \theta X_{t+1} \Leftrightarrow \frac{1}{X_{t+1} - 1} = \frac{1}{\beta \theta} \frac{1}{X_t - 1} - 1$

(2) $\lim_{t \rightarrow \infty} \beta^t \frac{k_{t+1}^\theta}{c_t} = \lim_{t \rightarrow \infty} \beta^t \frac{1}{\frac{k_t^\theta}{k_{t+1}^\theta} - 1} = \lim_{t \rightarrow \infty} \beta^t \frac{1}{X_t - 1} = 0$

(1) $\Leftrightarrow \frac{1}{X_{t+1} - 1} - \frac{\beta \theta}{1 - \beta \theta} = \frac{1}{\beta \theta} \left(\frac{1}{X_t - 1} - \frac{\beta \theta}{1 - \beta \theta} \right)$

$\Leftrightarrow \frac{1}{X_{t+1} - 1} = \frac{\beta \theta}{1 - \beta \theta} + \left(\frac{1}{\beta \theta} \right)^t \left[\frac{1}{X_0 - 1} - \frac{\beta \theta}{1 - \beta \theta} \right]$ or $X_t = \frac{1}{\beta \theta}$. Fixed point.

(2): $\lim_{t \rightarrow \infty} \beta^t \frac{1}{X_t - 1} = \lim_{t \rightarrow \infty} \frac{\beta^{t+1} \theta}{1 - \beta \theta} + \frac{1}{\theta^t} \left[\frac{1}{X_0 - 1} - \frac{\beta \theta}{1 - \beta \theta} \right] = 0$

$\Leftrightarrow X_t = \frac{1}{\beta \theta}$. Transversality $\Leftrightarrow$ Constant saving rule.

Policy: $k_{t+1} = \beta \theta k_t^\theta$

Method 2: Guess & Verify (Guess from policy).

Guess $k_{t+1} = \gamma k_t^\theta$: $\ln k_{t+1} = \theta \ln k_t + \ln \gamma \quad (1)$

$c_t = k_t^\theta - k_{t+1} = (1 - \gamma) k_t^\theta$: $\ln c_t = \theta \ln k_t + \ln(1 - \gamma)$

$\Leftrightarrow \beta^t \ln c_t = \theta \beta^t \ln k_t + \beta^t \ln(1 - \gamma)$

$\Leftrightarrow \sum_{t=0}^{\infty} \beta^t \ln c_t = \theta \underbrace{\sum_{t=0}^{\infty} \beta^t \ln k_t}_Q + \frac{\ln(1 - \gamma)}{1 - \beta}$

(1) $\Leftrightarrow \beta^{t+1} \ln k_{t+1} = \beta \theta \beta^t \ln k_t + \beta \cdot \beta^t \ln \gamma$

$\underbrace{\sum_{t=0}^{\infty} \beta^{t+1} \ln k_{t+1}}_{Q - \ln k_0} = \beta \theta \underbrace{\sum_{t=0}^{\infty} \beta^t \ln k_t}_Q + \beta \frac{1}{1 - \beta} \ln \gamma$

$$\Leftrightarrow Q = \frac{1}{1-\beta\theta} \ln k_0 + \frac{\beta}{1-\beta} \frac{1}{1-\beta\theta} \ln \gamma$$

$$\Rightarrow \sum_{t=0}^{\infty} \beta^t \ln C_t = \frac{\theta}{1-\beta\theta} \ln k_0 + \frac{1}{1-\beta} \ln(1-\gamma) + \frac{\beta}{1-\beta} \frac{1}{1-\beta\theta} \ln \gamma = V(k_0)$$

$$\Rightarrow V(k_{t+1}) = \frac{\theta}{1-\beta\theta} \ln k_{t+1} + \text{Const.}$$

Plug in Bellman: $V(k_t) = \max_{k_{t+1}} \{ \ln(k_t^\theta - k_{t+1}) + \beta V(k_{t+1}) \}$

FOC: $\frac{1}{k_t^\theta - k_{t+1}} = \beta V'(k_{t+1}) = \frac{\beta\theta}{1-\beta\theta} \frac{1}{k_{t+1}} \Rightarrow k_{t+1} = \beta\theta k_t^\theta$

Verify: $\gamma = \beta\theta$ is a constant.

Policy: $k_{t+1} = \beta\theta k_t^\theta$

Transition:

Euler: $C_t = \beta\theta C_{t-1} k_t^{\theta-1}$

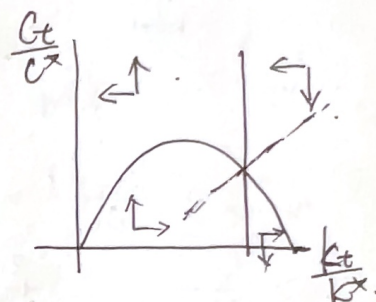
At SS: $C^* = \beta\theta C^* k^{*\theta-1}$

Budget: $C_t + k_{t+1} = k_t^\theta$

At SS: $C^* + k^* = k^{*\theta}$

$$\Rightarrow f: \begin{cases} \frac{C_t}{C^*} = \frac{C_{t-1}}{C^*} \cdot \left(\frac{k_t}{k^*}\right)^{\theta-1} \\ \frac{k_{t+1}}{k^*} = \left(\frac{k_t}{k^*}\right)^\theta - \frac{1-\beta\theta}{\beta\theta} \frac{C_t}{C^*} \end{cases}$$

With initial point $\begin{bmatrix} \frac{C_0}{C^*} = 1 \\ \frac{k_1}{k^*} \neq 1 \end{bmatrix}$



Locally stability: $J = \begin{bmatrix} \left(\frac{k_t}{k^*}\right)^{\theta-1} & (\theta-1)\left(\frac{k_t}{k^*}\right)^{\theta-2} \left(\frac{C_{t-1}}{C^*}\right) \\ -\frac{1-\beta\theta}{\beta\theta} & \theta\left(\frac{k_t}{k^*}\right)^{\theta-1} \end{bmatrix} \stackrel{\text{At SS}}{=} \begin{bmatrix} 1 & \theta-1 \\ -\frac{1-\beta\theta}{\beta\theta} & 1 \end{bmatrix}$

$$|J| = 1 + (\theta-1) \frac{1-\beta\theta}{\beta\theta} = 1 - (1-\theta) \frac{(1-\beta\theta)}{\beta\theta}$$

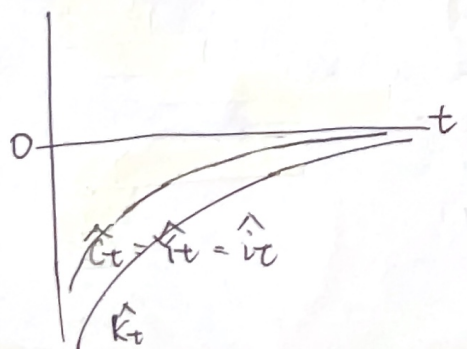
Eigenvalues $\lambda_1 = 1 - \sqrt{\frac{(1-\theta)(1-\beta\theta)}{\beta\theta}}$ $\lambda_2 = 1 + \sqrt{\frac{(1-\theta)(1-\beta\theta)}{\beta\theta}}$

$$x_1 = \begin{bmatrix} 1 \\ \sqrt{\frac{1-\beta\theta}{\beta\theta(1-\theta)}} \end{bmatrix}$$

Transition path.
Stable!

$$x_2 = \begin{bmatrix} 1 \\ -\sqrt{\frac{1-\beta\theta}{\beta\theta(1-\theta)}} \end{bmatrix}$$

unstable!



$$\hat{x}_t := \ln \frac{x_t}{x^*}$$

$$x_t = k_t^\theta \Rightarrow \ln \frac{x_t}{x^*} = \theta \ln \frac{k_t}{k^*}$$

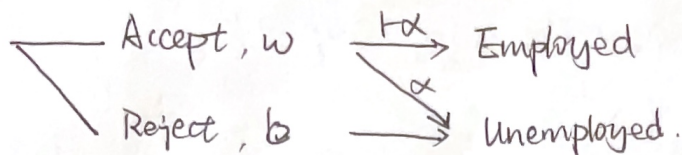
$$C_t = (1-\beta\theta) k_t^\theta \Rightarrow \ln \frac{C_t}{C^*} = \theta \ln \frac{k_t}{k^*}$$

$$\hat{c}_t = \beta\theta k_t^\theta \Rightarrow \ln \frac{\hat{c}_t}{\hat{c}^*} = \theta \ln \frac{k_t}{k^*}$$

Single-Side Search Model

Unemployed.

$w \sim F$ on $[0, B]$



Bellman: $V(w) = \max \{ w + \beta(1-\alpha)V(w) + \beta\alpha \mathbb{E}V, b + \beta \mathbb{E}V \}$.

Consider 2 states { employed with w : $V^e(w) = w + \beta(1-\alpha)V^e(w) + \beta\alpha \mathbb{E}V^u(w)$.

{ unemployed with w : $V^u(w) = \max \{ w + \beta(1-\alpha)V^e(w) + \beta\alpha \mathbb{E}V^u, b + \beta \mathbb{E}V^u \}$

$$\Rightarrow V^e(w) = \frac{w}{1-\beta(1-\alpha)} + \frac{\beta\alpha}{1-\beta(1-\alpha)} \mathbb{E}V^u.$$

$$\Rightarrow V^u(w) = \max \left\{ \frac{w}{1-\beta(1-\alpha)} + \frac{\beta\alpha}{1-\beta(1-\alpha)} \mathbb{E}V^u, b + \beta \mathbb{E}V^u \right\} \text{ non-decreasing on } w.$$

$$\exists \text{ reservation } w^*: \frac{w^*}{1-\beta(1-\alpha)} + \frac{\beta\alpha}{1-\beta(1-\alpha)} \mathbb{E}V^u = b + \beta \mathbb{E}V^u. \quad (1)$$

$$\text{and } V^u(w) = \begin{cases} \frac{w}{1-\beta(1-\alpha)} + \frac{\beta\alpha}{1-\beta(1-\alpha)} \mathbb{E}V^u & \text{if } w \geq w^* \\ b + \beta \mathbb{E}V^u = \frac{w^*}{1-\beta(1-\alpha)} + \frac{\beta\alpha}{1-\beta(1-\alpha)} \mathbb{E}V^u & \text{if else.} \end{cases} \quad (2)$$

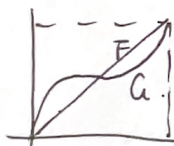
$$(1) \Leftrightarrow w^* = [1-\beta(1-\alpha)]b + \beta(1-\beta)(1-\alpha) \mathbb{E}V^u$$

$$(2) \Rightarrow \mathbb{E}V^u = \int_{w^*}^B \frac{w}{1-\beta(1-\alpha)} + \frac{\beta\alpha}{1-\beta(1-\alpha)} \mathbb{E}V^u dF(w) + \int_0^{w^*} \frac{w^*}{1-\beta(1-\alpha)} + \frac{\beta\alpha}{1-\beta(1-\alpha)} \mathbb{E}V^u dF(w).$$

$$\Rightarrow (1-\beta) \mathbb{E}V^u = \int_{w^*}^B (1-F(w)) dw + w^*$$

$$(1) \& (2) \Rightarrow w^* = b + \frac{\beta(1-\alpha)}{1-\beta(1-\alpha)} \int_{w^*}^B (1-F(w)) dw.$$

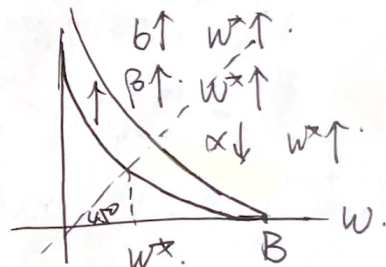
$$\Leftrightarrow w^* = [1-\beta(1-\alpha)]b + \beta(1-\alpha) \left[\mathbb{E}w + \int_0^{w^*} F(w) dw \right].$$



Mean preserving spread. $w^* \uparrow$.

$$F \rightarrow G. \quad \forall x. \int_0^x (G(x) - F(x)) dx \geq 0.$$

Probability to get employed in each period: $P = 1 - F(w^*)$



Search Equilibrium Model (Diamond 1982)

$$V_0 = -\gamma_0 + \beta \left(\alpha_0 \int \max \{ V_1 - c, V_0 \} dF(c) + (1-\alpha_0) V_0 \right).$$

$$V_1 = -\gamma_1 + \beta \left(\alpha_1 (u + V_0) + (1-\alpha_1) V_1 \right).$$

Reservation K : $K = V_1 - V_0$: $\max \{ V_1 - c, V_0 \} = \begin{cases} V_1 - c & \text{if } c < K \\ V_0 & \text{otherwise} \end{cases}$

$$V_0 = -\gamma_0 + \beta \alpha_0 \int_0^K \underbrace{(V_1 - c)}_{=V_0+K} dF(c) + \beta \alpha_0 \int_0^K V_0 dF(c) + \beta (1-\alpha_0) V_0.$$

$$= -\gamma_0 + \beta \alpha_0 \int_0^K (K - c) dF(c) + \beta V_0.$$

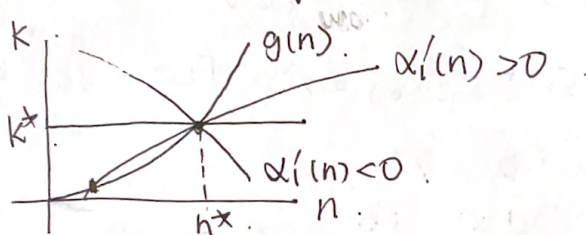
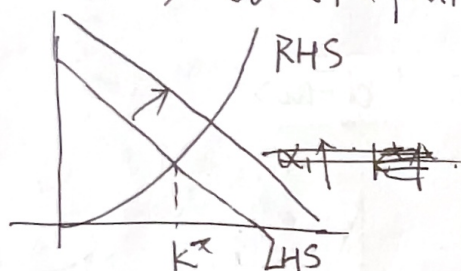
$$\Rightarrow (1-\beta) V_0 = -\gamma_0 + \beta \alpha_0 \int_0^K (K - c) dF(c). \quad (1)$$

$$V_1 = -\gamma_1 + \beta \alpha_1 u + \beta V_1 - \beta \alpha_1 \underbrace{(V_1 - V_0)}_{=K}$$

$$\Rightarrow (1-\beta) V_1 = -\gamma_1 + \beta \alpha_1 u - \beta \alpha_1 K.$$

$$\Rightarrow (1-\beta) V_0 = (1-\beta) (V_1 - K) = -\gamma_1 + \beta \alpha_1 u - (1-\beta + \beta \alpha_1) K. \quad (2)$$

$$(1) \& (2) \Rightarrow \gamma_0 - \gamma_1 + \beta \alpha_1 u - (1-\beta + \beta \alpha_1) K = \beta \alpha_0 \int_0^K (K - c) dF(c).$$



Market clearing: Production: $(1-n) \alpha_0 F(K) = n \alpha_1$. Exchange (consumption)

$$\Rightarrow n = \frac{\alpha_0 F(K)}{\alpha_0 F(K) + \alpha_1} \Rightarrow K = g(n).$$

Consider externalities: $\alpha_1(n)$: $\alpha_1'(n) < 0$. Congestion. negative externality.
 > 0 . Thick market. positive externality.

$$\begin{cases} \gamma_0 - \gamma_1 + \beta \alpha_1(n) u - (1-\beta + \beta \alpha_1(n)) K = \beta \alpha_0 \int_0^K (K - c) dF(c) \\ n = \frac{\alpha_0 F(K)}{\alpha_0 F(K) + \alpha_1(n)} \end{cases}$$

Social planner $\begin{cases} K_{ss}^* > K^* & \text{if } \alpha_1'(n) > 0 \\ K_{ss}^* < K^* & \text{if } \alpha_1'(n) < 0 \end{cases}$ Consider $K_{ss} = K + \delta$.

$$n(\delta) = \frac{\alpha_0 F(K + \delta)}{\alpha_0 F(K + \delta) + \alpha_1(n(\delta))}$$

$$\begin{cases} (1-\beta) V_0(\delta) = -\gamma_0 + \beta \alpha_0 \int_0^{K+\delta} (K + \delta - c) dF(c) \\ (1-\beta) V_1(\delta) = -\gamma_1 + \beta \alpha_1(n(\delta)) u - \beta \alpha_1(n(\delta)) (K + \delta) \end{cases}$$